

Recycling of aggregates in the World



Working for Sustainable Development

NATIONAL STRATEGIC PROJECT

**On the Collection of Demolition Rubble and Glass as a Model of Circular
Economy**

ROMANIA 2027-2040

Prepared by: Green Mineral SRL (Italy) - International Operations

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Project Scope: Romanian municipalities

Partners: Green Mineral Italy | Alseco SRL (Financial Partner) | Romanian Public Entities

Timeline: 20-Year National Programme (Phase 1 launch: 2027-2028)

Patent: Zero Process Waste (MISE Patent No. 102017000077948 of 08-01-2020)

Funding Request: EU programmes, national and municipal sources, EIB and EBRD facilities

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WORLD BANK GROUP
EIB SUPPORTED
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1. LETTER OF INTRODUCTION

RO PUBLIC-PRIVATE PARTNERSHIP (PPP) PROPOSAL

SUBJECT: Establishment of a public-municipal circular economy model for construction and demolition waste, separately collected glass and recycled construction granulates in Romania.

Dear Honorable Minister,

GREEN MINERAL is an Italian multinational company specializing in the recovery and recycling of inert materials. The organization has established proven Circular Economy Models by developing and implementing integrated supply chain projects worldwide, primarily across Europe, North Africa, and Asia. These operations are fully aligned with the European Waste Framework Directive (WFD) 2008/98/EC, strictly promoting sustainable waste management according to the waste hierarchy and supporting the global goal of 70% waste recovery.

Green Mineral specializes in recycling construction waste from demolitions using proprietary patented technologies (MISE Patent No. 102017000077948, dated 08-01-2020). This advanced system enables a **Zero Process Waste** discharge, transforming all construction debris into high-quality commercial products such as certified aggregates, tiles, concrete components, and prefabricated blocks. This process greatly reduces landfill dependency and environmental pollution while generating local employment and stimulating regional economies.

Following the successful implementation of several pilot projects across Europe, Green Mineral proposes a Public-Private Partnership (PPP) program for Romania. This initiative is aimed at post-war reconstruction through sustainable waste management, high-technology recycling systems, and the creation of local green jobs.

Green Mineral plans to establish regional waste recycling hubs and selective collection centers in key Romanian municipalities, integrating circular economy principles into the country's reconstruction process. This framework is fully aligned with **Romania's National Waste Management Strategy (NWS)**, the **European Green Deal**, and the

United Nations Sustainable Development Goals (UN-SDGs 1, 3, 9, 11, and 12).

The company's specialized *Riccitiello's* and *ZERO WASTE CONSTRUCTION* programmes will also target end-of-life tires (ELTs) and mixed industrial waste, converting them into eco-friendly building materials. These large-scale infrastructure projects can be sustainably financed through EU programmes, green funds and loans from institutions such as the World Bank Group (WB), the European Investment Bank (EIB), and the European Bank for Reconstruction and Development (EBRD).

With Best Regards,

Nando Cocco

CEO & Founder, Green Mineral SRL

PROJECT OVERVIEW:

“ROMANIA NATIONAL CIRCULAR CONSTRUCTION MATERIALS PROGRAMME 2027–2040”

Romania faces significant environmental and infrastructure challenges arising from illegal landfilling, uncontrolled waste disposal, and the limited high-quality recycling of construction and demolition waste.

Under Romania’s EU Accession Treaty, 101 non-compliant landfills were required to be closed and rehabilitated by 16 July 2017. Although 92 sites have since been closed and rehabilitated, nine remain outstanding, prompting the European Commission to issue a reasoned opinion for non-compliance with the Landfill Directive and Waste Framework Directive.

At the same time, Romania has achieved approximately 70–73% material recovery of construction and demolition waste, while actual recycling has remained considerably lower. This gap indicates that substantial quantities are still disposed of, used for low-value recovery or managed without being converted into tested, marketable secondary construction materials.

The proposed programme establishes a coordinated national circular-economy model for the separate collection, traceable transport, processing and productive reuse of demolition rubble and glass. Working through municipalities and publicly controlled sanitation companies, the programme will support the production of certified recycled aggregates for roads, platforms, trench backfilling and other public works. Implemented progressively from municipal pilots to regional and national replication, the 2027–2040 programme aims to reduce illegal dumping, divert mineral waste from landfills, strengthen environmental compliance and create a sustainable Romanian market for recycled construction materials.

ROMANIA NATIONAL CONTEXT

Romania's C&D market is commercially established at the collection and disposal stage but less mature at the certified recycled-product stage. The 2025 draft revised National Waste Management Plan reports approximately 161 authorised operators receiving or treating C&D waste and identifies continued disposal in municipal landfills where transport economics make this attractive. At the same time, Romania's unresolved landfill-closure obligations have moved from administrative warning to judicial enforcement at EU level.

STRATEGIC GAP

Reported material recovery is not the same as a functioning circular materials market. Backfilling and other recovery can help meet the 70% target while producing limited marketable aggregate. The programme should therefore track clean feedstock, certified product yield, tonnes sold, virgin aggregate displaced and actual use in public works.

EU INFRINGEMENT STATUS AND PROGRAMME BOUNDARY

On 7 February 2024, the European Commission issued a reasoned opinion in INFR (2020)2276 after 92 of 101 relevant landfills had been closed and rehabilitated but nine remained outstanding. On 29 January 2026, the Commission referred Romania to the Court of Justice of the European Union because those nine sites were still not rehabilitated. A municipal circular-materials programme can support compliance by providing a lawful collection and recovery route, reducing fresh inflows to non-compliant sites, preventing recontamination and documenting diversion. It cannot replace site-specific closure design, contaminated-land investigation, capping, leachate or gas controls, rehabilitation works, aftercare or regulatory reporting.

(Source: Romania Ministry of Environment, draft revised National Waste Management Plan, 8 August 2025, C&D chapter; European Commission infringement decisions of 7 February 2024 and 29 January 2026).

C&D RECYCLING PERFORMANCE AND THE QUALITY GAP

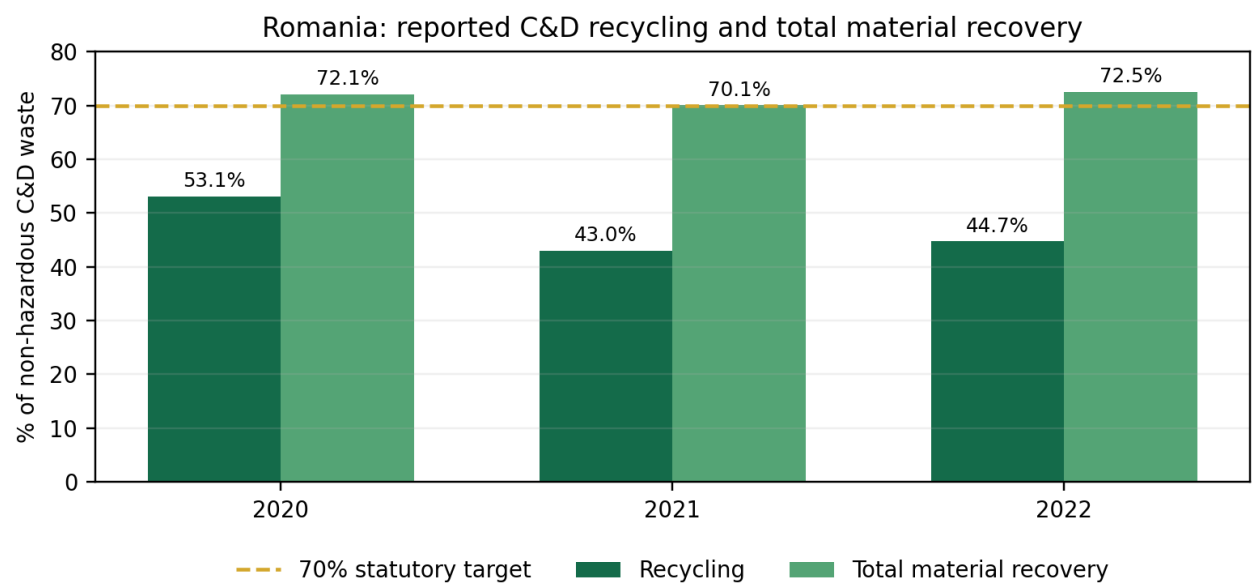


Figure 1.1: Romania C&D recycling and total material recovery).
(Source: Reported rates in the 2025 draft revised National Waste Management Plan.)

INTERPRETATION

Romania exceeded the 70% total material-recovery line in each reported year, but recycling fell from 53.1% in 2020 to 43.0% in 2021 and recovered only slightly to 44.7% in 2022. This difference is the principal market opportunity: develop facilities, standards and public demand that convert mixed rubble and backfilling-oriented recovery into traceable recycled products.

HOW WE WORK

Our method is based on three distinct process phases, with the ability to employ a wide range of skilled workers.

Green Mineral was awarded a zero-waste process patent by the Italian Ministry of Economic Development, MISE, Patent No. 102017000077948, dated January 8, 2020.

WHY WE ARE INNOVATIVE

We are the first in the world to use an innovative method to control the flow of demolition debris, from the demolition site to the recovery plant.

Our goal is to eliminate the proliferation of illegal landfills, reduce environmental impact, and lower cleanup costs that are currently borne by the community.

2. DISTRIBUTION - PROJECT PHASES & EMPLOYMENT PLAN

The Romania National Circular Construction Materials Programme 2027–2040 proposes a hub-and-spoke network of municipal collection centres, inter-municipal processing hubs, recycled-concrete product units, and advanced construction, demolition and glass-recycling facilities. Romania comprises 319 urban localities, including 103 municipalities and 216 towns, together with 2,862 communes. Consequently, the programme should not establish an independent processing plant in every locality. Large municipalities would host integrated facilities, while smaller towns and communes would be served through shared collection centres, mobile processing campaigns and regional recycling hubs.

The national chain factories and collection centers allocation plan encompasses total direct and indirect workforce deployment across three distinct operational phases:

1. **Phase 1:** Construction and setup of selective waste collection centers (utilizing fixed-term specialized labor).
2. **Phase 2:** Establishment of concrete product manufacturing plants (operational production phase).
3. **Phase 3:** Integration of advanced C&D and glass recycling plants, including high-precision sorting and automated bagging operations.

Workforce distribution model scaled for large cities and municipalities ranging from 1,000 to 3,000,000 inhabitants:

#	City/Municipal Cluster	County/Region	Population	Phase 1	Phase 2	Phase 3	Collection C.	Total Workers
01	București	Bucharest Municipality	1.716.961	6	4	2	12	220
02	Ilfov Metropolitan Cluster	Ilfov	542.704	4	2	1	8	140
03	Cluj-Napoca	Cluj	286.598	3	2	1	6	120
04	Iași	Iași	271.692	3	2	1	6	120
05	Constanța	Constanța	263.688	3	2	1	6	120
06	Timișoara	Timiș	250.849	3	2	1	6	120
07	Brașov	Brașov	237.589	3	2	1	5	105
08	Craiova	Dolj	234.140	3	2	1	5	105

#	City/Municipal Cluster	County/Region	Population	Phase 1	Phase 2	Phase 3	Collection C.	Total Workers
09	Galati	Galati	217.851	3	2	1	5	105
10	Oradea	Bihor	183.105	2	2	1	5	90
11	Ploiesti	Prahova	180.540	2	2	1	5	90
12	Braila	Braila	154.686	2	2	1	5	90
13	Arad	Arad	145.078	2	1	1	4	75
14	Pitesti	Arges	141.275	2	1	1	4	75
15	Bacau	Bacau	136.087	2	1	1	4	75
16	Sibiu	Sibiu	134.309	2	1	1	4	75
17	Targu Mures	Mures	116.033	2	1	1	4	75
18	Baia Mare	Maramures	108.759	2	1	1	4	75
19	Buzau	Buzau	103.481	2	1	1	4	75
20	Ramnicu Valcea	Valcea	93.151	1	1	1	3	50
21	Satu Mare	Satu Mare	91.520	1	1	1	3	50
22	Botosani	Botosani	90.010	1	1	1	3	50
23	Suceava	Suceava	84.322	1	1	1	3	50
24	Drobeta-Turnu Severin	Mehedinti	79.865	1	1	1	3	50
25	Piatra-Neamt	Neamt	79.679	1	1	1	3	50
26	Bistrita	Bistrita-Nasaud	78.877	1	1	1	3	50
27	Targu Jiu	Gorj	73.545	1	1	—	3	40
28	Targoviste	Dambovita	66.965	1	1	—	3	40
29	Focșani	Vrancea	66.648	1	1	—	3	40
30	Tulcea	Tulcea	65.624	1	1	—	3	40
31	Alba Iulia	Alba	64.227	1	1	—	3	40
32	Slatina	Olt	63.487	1	1	—	3	40
33	Vaslui	Vaslui	63.035	1	1	—	3	40
34	Reșita	Caras-Severin	58.393	1	1	—	3	40
35	Călărași	Călărași	58.211	1	1	—	3	40
36	Giurgiu	Giurgiu	54.551	1	1	—	3	40
37	Deva	Hunedoara	53.113	1	1	—	3	40
38	Zalău	Sălaj	52.359	1	1	—	3	40
39	Sfântu Gheorghe	Covasna	50.080	1	1	—	3	40
40	Slobozia	Ialomita	41.550	1	—	—	2	25
41	Alexandria	Teleorman	40.390	1	—	—	2	25
42	Miercurea Ciuc	Harghita	34.484	1	—	—	2	25
43	Other urban localities: 50,000–99,999	National, excluding Ilfov cluster	2 UATs / 102.932	2	2	—	6	70

#	City/Municipal Cluster	County/Region	Population	Phase 1	Phase 2	Phase 3	Collection C.	Total Workers
44	Other urban localities: 20,000–49,999	National, excluding Ilfov cluster	40 UATs / 1.143.374	14	5	3	80	600
45	Other urban localities: 10,000–19,999	National, excluding Ilfov cluster	80 UATs / 1.110.655	16	4	2	160	800
46	Other urban localities: below 10,000	National, excluding Ilfov cluster	148 UATs / 950.069	15	—	—	148	592
47	Large communes: 20,000 and above	National, excluding Ilfov cluster	4 communes / 123.553	2	1	—	4	40
48	Communes: 10,000–19,999	National, excluding Ilfov cluster	31 communes / 384.495	4	1	—	31	155
49	Communes: 5,000–9,999	National, excluding Ilfov cluster	339 communes / 2.157.329	12	2	1	113	452
50	Communes: 2,000–4,999	National, excluding Ilfov cluster	1.570 communes / 4.905.570	20	3	1	314	942
51	Communes: below 2,000	National, excluding Ilfov cluster	886 communes / 1.246.327	10	—	—	148	296
TOTAL ROMANIA		National programme	19,053,815	169	71	34	1.172	6.842

3. GEOGRAPHICAL COVERAGE & TERRITORIAL ELEVATION MAP

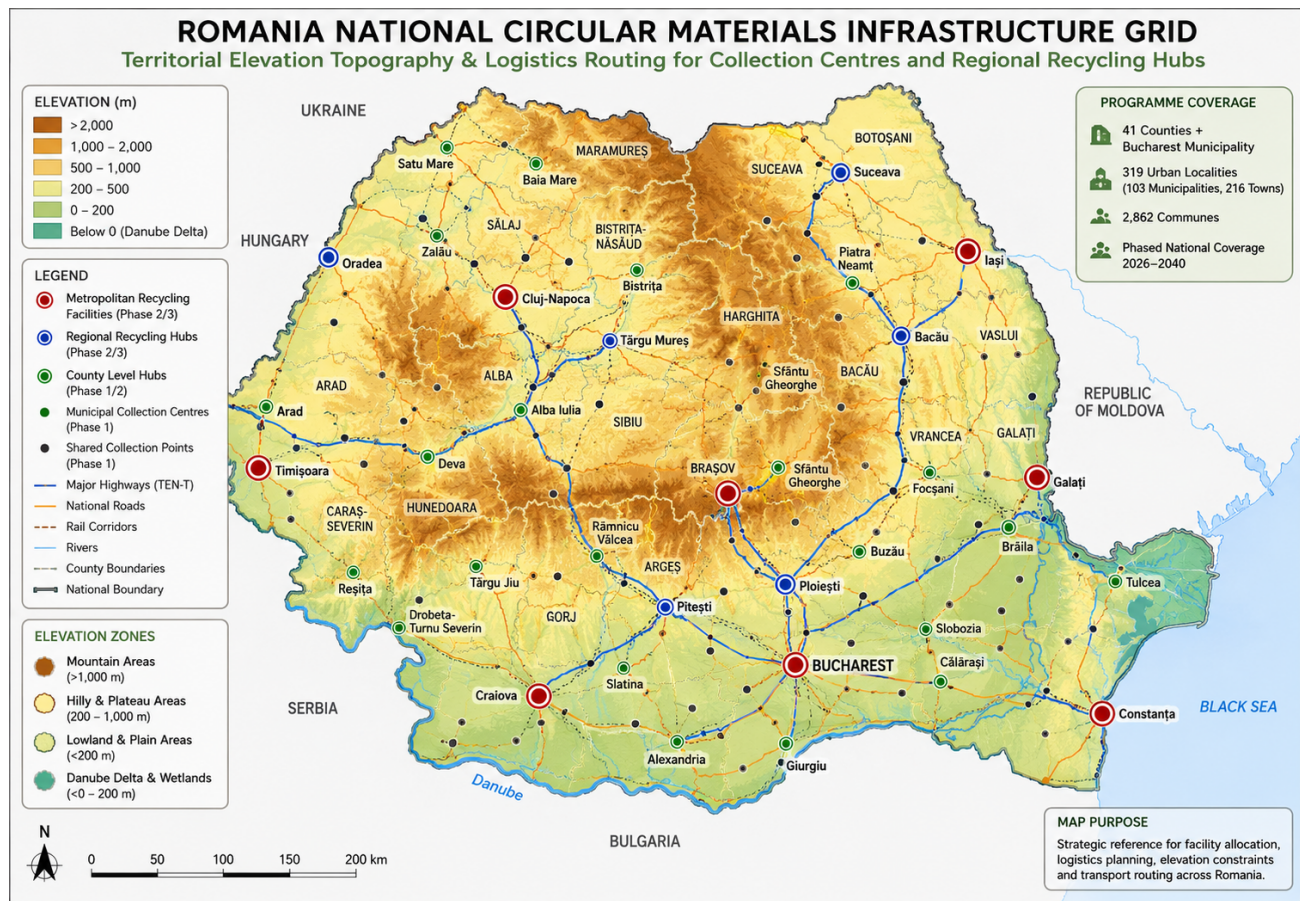


Figure 3.1 : Romania National Infrastructure Grid and Elevation Topography mapping for logistics routing (GRID-Arendal Data Integration).

The Romania National Circular Construction Materials Programme will provide phased coverage across all 41 counties and the Municipality of Bucharest. The territorial model will combine metropolitan recycling facilities, county-level processing hubs, municipal collection centres and shared collection points serving smaller towns and rural communes.

Facility location and logistics planning will take account of Romania's varied terrain, including the Carpathian Mountains, Transylvanian Plateau, Moldavian and Dobrogea regions, the Danube corridor, the Romanian Plain and the Black Sea coast. Elevation, road accessibility, settlement density, transport distance, construction activity, available public land and proximity to public-works markets will guide the selection of collection centres and regional processing hubs.

4. CERTIFICATE - PATENT SPECIFICATIONS

Green Mineral operates under a strictly protected proprietary industrial invention patent, certified by the Italian Ministry of Economic Development (MISE). This ensures total compliance with international industrial standards for secondary raw material manufacturing.



Ministry of Economic Development
General Directorate for the Protection of Industrial Property

Italian Patent and Trademark Office

PATENT CERTIFICATE FOR INDUSTRIAL INVENTION

This patent is granted for the invention subject of the application:

N. 102017000077948

TITLE HOLDER:

- GREEN MINERAL 90.0%
- Cocco Nando 10.0%

DOMICILE: Nando COCCO
Via Moreolense SNC
03017 Morolo

INVENTOR: Cocco Nando

TITLE: RECYCLING INERT MATERIALS FROM DEMOLITION OBTAINING ZERO WASTE

CLASSIFICATION: B03B

FILING DATE: 12/07/2017

Rome, 08/01/2020

The Division Manager

Loredana Guglielmetti

5. IMPLEMENTATION AND PARTNERSHIPS

The operational rollout of the Romanian National Programme leverages consolidated joint-venture frameworks and deep multi-sector professional expertise:

NCL GROUP SRL

Operating continuously within the environmental remediation, industrial waste management, and site decontamination sectors since 1998. Bringing over 25 years of corporate growth and highly specialized operational assets, NCL Group maintains a robust roster of senior civil engineers, heavy plant operators, regulatory compliance officers, and logistical dispatch personnel.

ALSECO SRL

An established global trading house specializing in scrap-derived commodities and international alternative fuels brokerage. Alseco's core competencies lie in managing complex cross-border logistics, supply-chain quality control metrics, and localized end-market distribution channels for processed rubber compounds, scrap polymers, paper, and non-ferrous metals.



ALSECO Srl operates in the international scrap derived goods trading.

Thanks to its "know-how" coming from more than twenty years of experience of its founders in the production and exportation of alternative fuels (products made from the tire scrap recycling process, such as rubber compound and scrap plastics, scrap paper, and scrap metal), ALSECO is well known as a successful company in the exportation of the above mentioned goods.

The careful choice of raw materials, the collaboration with renown producers and logistic companies, the qualitative controls in every step of production, together with the deep knowledge of the markets of final destination, ALESCO SRL meets the highest quality goods standards and efficiency in delivery.

Figure 5.1: Alseco global distribution channels and material validation matrix.

6. CORPORATE PROFILE: NCL GROUP S.R.L.

As documented in the supporting technical dossier, NCL GROUP S.r.l. represents the foundational operational pillar of this joint infrastructure programme. Operating under strict European compliance frameworks, including the adoption of the ethical code and organizational models pursuant to Italian Legislative Decree 231/01 (Modello ex D.Lgs. 231/01), the group guarantees absolute transparency and governance across complex public-private reconstruction tenders.

THE COMPANY & INDUSTRIAL EVOLUTION

The industrial legacy spans over 40 years, initiated by Ormello Cocco who established the initial nucleus of what has evolved into one of the most critical infrastructure and aggregate raw material entities within the Lazio region. The corporate asset footprint includes three active limestone quarries located within the Province of Frosinone, driving large-scale operations in earthmoving, technical excavation, complex structural demolition, logistics transportation, and secondary raw material recycling.

The continuous structural and technological adaptation spearheaded by the Cocco family has maintained a superior professional edge, resulting in a diversified structure composed of five independent, highly specialized group companies operating synergistically.

CORE OPERATIONAL SERVICES

- **High-Elevation Controlled Demolition:** Executing heavy demolition of reinforced concrete and complex structural steel frameworks utilizing ultra-advanced mechanized techniques.
- **Raw Material Extraction & Processing:** Production of high-grade washed aggregates directly from proprietary limestone reserves.
- **Environmental Remediation & Marine Works:** Large-scale site reclamation, dredging of industrial canals, reservoir restoration, and comprehensive topographic and geological subsoil modeling.
- **Circular Economy Deployment:** Integration of specialized tire scrap recycling lines and advanced construction and demolition (C&D) waste recovery systems under the proprietary Green Mineral industrial process.

7. FACTORIES AND COLLECTION CENTERS

To support the target municipalities across Romania, Green Mineral will systematically deploy specialized processing hubs and distributed drop-off infrastructure optimized across three synchronized development phases:

OPERATIONAL PHASE	PLANT FUNCTIONAL DESCRIPTION	CAPACITY & SITE SPECIFICATIONS
Phase 1: Fixed Processing Plant	Primary mechanical crushing, automated sorting, magnetometric separation, and precise qualitative classification of incoming construction and demolition rubble. Permanent heavy installation outfitted with electronic weighbridges, structured material discharge zones, and integrated Line A & B processing arrays.	Processing Capacity: 300,000 to 1,500,000 tonnes/year (scaled to municipal size). Site Allocation: 15,000 – 20,000 m².
Phase 2: Concrete Products Unit	Secondary industrial processing and value-add manufacturing. Direct transformation of recycled granulates into certified structural concrete blocks, modular pavement elements, and heavy retaining wall units.	Additional Site Footprint: 5,000 – 10,000 m². Output: High-demand certified structural units. Achieves true zero secondary waste emissions.
Phase 3: Automated Bagging Line	Tertiary refinement and processing of residual fines. Fully automated bagging, palletizing, and serialization of standardized retail-grade and industrial-grade granulates intended for general distribution markets.	Market Destination: Pre-packed calibrated fractions for hardware retail networks, DIY distributors, and local renovation works.
Distributed Mini Collection Centres	A hyper-local network of secure, urban collection hubs designed for efficient C&D waste drop-off by private citizens, municipal maintenance crews, and independent small contractors.	Standard Unit Size: 500 – 2,000 m². Manning: 2–4 workers per hub. Logistically linked via scheduled truck shuttles to primary regional plants.

INITIAL PHASE 1 NATIONAL INFRASTRUCTURE FRAMEWORK

Based on the indicative territorial distribution and employment model, the first-wave infrastructure programme will cover 21 priority municipalities and metropolitan clusters across Romania. The proposed infrastructure framework comprises:

- A. **Fixed Processing Plants (Phase 1):** ~ Approximately 53 fixed or modular installations for receiving, weighing, inspecting, separating and preparing construction and demolition waste and glass for further processing.
- B. **Concrete Products Manufacturing Units (Phase 2):** ~ 35 integrated production units
- C. **Automated Bagging Lines (Phase 3):** ~22 industrial lines.
- D. **Mini Collection Centres:** ~ 108 distributed collection and transfer centres
- E. **Total Projected Infrastructure Footprint:** ~ **218 infrastructure components** across the Phase 1 national roll-out.

8. DIRECT AND INDIRECT BENEFITS TO MUNICIPALITIES

DIRECT MACRO-ECONOMIC BENEFITS

- A. **EU Standardization:** Implementation of sustainable, sound waste management protocols in full alignment with EU directives.
- B. **Landfill Life Extension:** Massive reduction of inert volumes entering municipal landfills, extending the operational lifecycle of urban solid waste (RSU) discharge infrastructure.
- C. **Employment & Tech Transfer:** Generation of hundreds of stable, local green-economy job opportunities and introduction of high-tech recycling systems.
- D. **Fiscal Direct Savings:** Provision of certified high-grade recycled granulates at **10% to 15% below traditional quarry market prices** for public infrastructure tenders.

INDIRECT SOCIO-ENVIRONMENTAL BENEFITS

- A- **Eradication of Illegal Dumping:** Systematic elimination of unauthorized dumping grounds, addressing social and ecological hazards in war-affected municipal zones.
- B- **Decentralized Transit Routing:** Recycling debris within its municipal origin point significantly minimizes material transport paths, leading to reduced fuel consumption and lower carbon emissions.
- C- **Geological Preservation:** Mitigates aggressive raw material extraction from quarries, lowering the ecological impact on natural landscapes and local water tables.

Benefit:

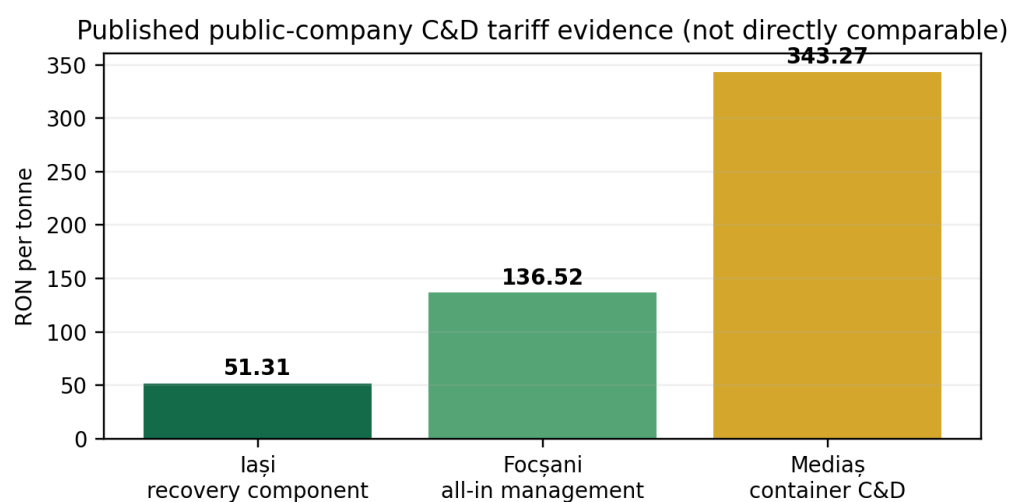
The municipality benefits only when the recycled product is actually used or sold. Stockpiled aggregate is not a completed circular-economy outcome.

RECOMMENDED DOSSIER STATEMENT

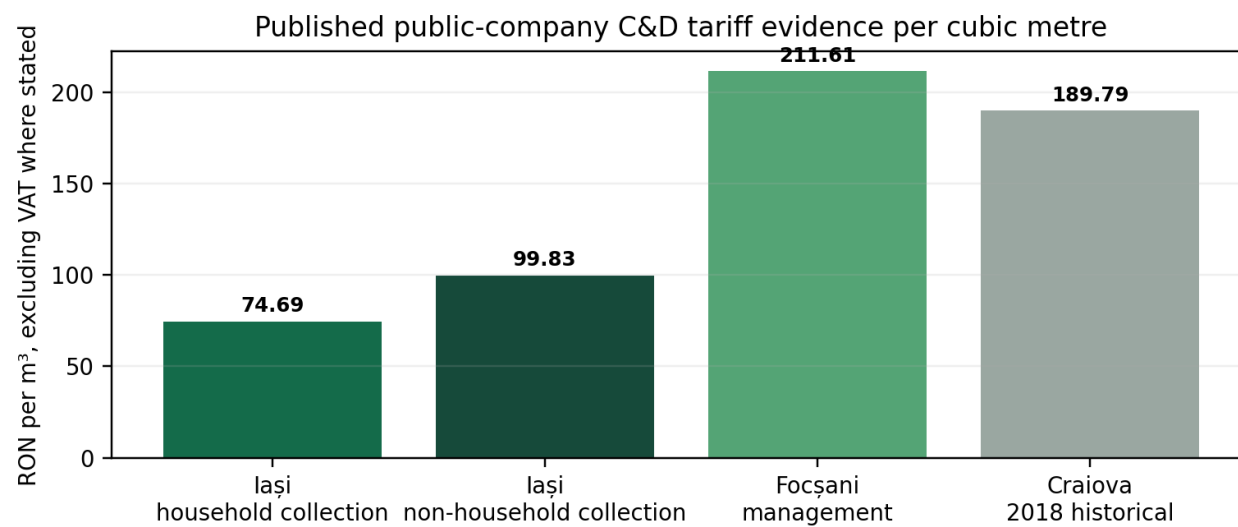
Official national records indicate **approximately 1.23 million tonnes** of construction and demolition waste in Romania in 2022. Region-specific registered quantities remain subject to extraction from the INSSE TEMPO database and confirmation by the competent environmental and local authorities. No unverified regional allocation is presented as official data.

9. PUBLIC C & D TARIFF COMPARISON

ID	Official body / file	Evidence used	URL / location
M 1	Craiova Municipality	2025 record identifying the municipality as sole shareholder	https://www.primariacraiova.ro/uploads/articole/attachments/67b855d849ac6848172347.pdf
M 2	Focșani Municipality	Public-company ownership report and HCL 254/2024 tariff	https://www.focsani.info/
M 3	ECO-SAL S.A.	100% public ownership and municipal shareholders	https://eco-sal.ro/conducerea-s-c-eco-sal-s-a/
M 4	ECO-SAL S.A.	C&D container rules and published rates	https://eco-sal.ro/comanda-container/
M 5	Botoșani Municipality / Urban Serv	Public ownership and sanitation regulation	https://www.primariabt.ro/ ; https://www.urbanserv.ro/
M 7	Salubrizare Sector 5 S.A.	Public sanitation and renovation-waste service information	https://salubrizare5.ro/



Different service scopes, dates and VAT bases. Use only for screening; request current quotations before procurement.



Craiova is a historical public benchmark and must not be treated as a current tariff.

10. PLANT TYPE (WITHDRAWALS / TREATING WASTES)

The standard industrial configuration of a permanent Green Mineral plant achieves a strict zero-waste operating lifecycle through the separate parallel operation of two distinct processing streams:

DUAL PROCESSING STREAM SEPARATION

Line A Architecture: Engineered specifically for the continuous processing, screening, and recovery of mixed building structures, mineral elements, and heterogeneous C&D fragments classified under the following European Waste Catalogue (EWC) codes:

- **EWC 17.01.02:** Inert waste components consisting exclusively of brick materials.
- **EWC 17.01.03:** Ceramic tiles, structural tiles, and high-density ceramic components.
- **EWC 17.01.07:** Heterogeneous mixtures of concrete, bricks, tiles, and ceramic fragments.
- **EWC 17.09.04:** Mixed construction and demolition waste fragments lacking hazardous constituents.
- **EWC 17.05.04:** Non-hazardous natural soils and mineral stones.
- **EWC 17.05.06 & EWC 17.08.02:** Non-hazardous dredged materials and clean gypsum building materials.

Line B Architecture: A dedicated line configured for high-purity glass separation, optical sorting, and the processing of clean monolithic concrete structural components and non-hazardous bituminous materials:

- A. **EWC 17.01.01:** Clean concrete structures and reinforced concrete fractions.
- B. **EWC 17.02.02:** Glass fragments, structural glazing, and container fractions.
- C. **EWC 17.03.02:** Non-hazardous bituminous mixtures and compatible industrial paving residues.

CORE PLANT COMPONENT MATRIX

PLANT COMPONENT	FUNCTIONAL SPECIFICATION & OPERATIONAL PARAMETERS
Weighbridge & Reception	All incoming transport fleets carrying C&D fractions undergo rigorous electronic weighing, automated classification, and complete statutory documentation mapped per EWC code prior to processing site discharge.

PLANT COMPONENT	FUNCTIONAL SPECIFICATION & OPERATIONAL PARAMETERS
Line A - Primary Crusher	High-torque mechanical jaw crusher integrated with multi-deck vibrating screening plants to process complex bricks, mixed concrete rubble, and multi-component masonry.
Line B - Glass & Concrete	Specialized crushing, air-classification, and optical sorting arrays designed for glass cullet processing, alongside high-capacity milling for pure concrete sorting.
Phase 3 Residue Treatment	Tertiary mechanical and physical processing of residual fine particles and mixed binding materials, preventing any bypass to landfill and fulfilling the zero-waste engineering mandate.
Dust & Water Management	Fully enclosed acoustic and material processing hulls, automated high-pressure water spray dust suppression systems, and closed-loop sediment settling and water recovery ponds.

11. PRODUCTION OF GRANULATES AND REUSE FRAMEWORK

Once processed through Green Mineral's advanced technological arrays, structural demolition aggregates bypass municipal landfill dumping completely, returning directly to the regional construction and infrastructure commercial markets as certified added-value products.

CALIBRATED GRANULATE OUTPUT MATRIX

PRODUCT GRADE	TECHNICAL PRODUCT DESCRIPTION	APPROVED ENGINEERING & CONSTRUCTION APPLICATIONS
Class 0–4 mm	Fine recycled mineral aggregate fraction.	High-density cement mixtures, road sub-base leveling layers, infrastructure pipe bedding, and technical floor screeds.
Class 4–8 mm	Medium recycled aggregate fraction.	Structural concrete production (approved for partial substitution matrices) and specialized environmental drainage layers.
Class 8–16 mm	Coarse recycled mineral aggregate fraction.	Structural concrete mixes (supporting up to 30% primary aggregate substitution), heavy road base engineering, and embankments.
Class 16–32 mm	Large structural aggregate fraction.	Heavy embankment stabilization for roads, massive structural foundations, deep drainage fields, and backfill for civil structures.
Recycled Glass Cullet	Highly calibrated and sorted processed glass fraction.	Primary glass manufacturing supply (furnace cullet), advanced asphalt road surfacing modifiers, and architectural decorative aggregates.
Bagged Granulate	Retail-grade pre-packed fractions in certified high-durability bags.	Direct commercial supply channels targeting hardware retail networks, DIY construction, and localized residential renovation works.

REGULATORY FRAMEWORK INTEGRATION FOR PUBLIC WORKS

To successfully integrate these secondary materials into large-scale municipal and national reconstruction programs, the implementation strategy incorporates a harmonized regulatory framework. This aligns with long-standing European procurement procedures, including Green Public Procurement (GPP) protocols active across Italy, which legally mandate the inclusion of specified percentages of recycled aggregates inside public works contracts:

- A. **Transport Infrastructure:** Mandatory integration within road construction, rehabilitation projects, and base/sub-base structural layers.
- B. **Structural Engineering:** Foundation backfilling for large-scale residential housing and commercial zones.
- C. **Civil Works:** Construction of retaining walls, environmental embankments, and public infrastructure drainage.
- D. **Precast Manufacturing:** Direct feed for municipal production of concrete blocks, standard curbs, and modular pavement elements.

12. PRICE ON INERT WASTE / GLASS

The financial equilibrium and long-term scalability of the Green Mineral Romania National Circular Construction Materials Programme will be supported by two complementary and comparatively resilient revenue streams:

1. **Gate / Tipping Fees:** Tariffs charged to households, construction companies, public institutions, contractors and other waste generators delivering construction and demolition waste to municipal collection centres or processing facilities.
2. **Secondary Raw Material Sales:** Commercial sale of certified recycled aggregates, separated metals, processed glass, calibrated mineral fractions and precast construction products.

NATIONAL ECONOMIC AND REGULATORY PRICE SIGNAL

Romania applies a circular-economy contribution of **RON 160 per tonne** to municipal waste and construction and demolition waste sent for disposal. This contribution increases the full cost of landfilling and strengthens the economic case for separate collection, recycling and productive reuse.

IAȘI MUNICIPAL TARIFFS

Salubris S.A. publishes tariffs approved through Iași Local Council Decisions No. 138 and 139 of 28 April 2025, effective from 1 May 2025. The published tariffs are expressed exclusive of VAT.

SERVICE CATEGORY	OFFICIAL TARIFF
Collection and transport of household renovation waste	RON 74.69/m ³ , excluding VAT
Collection and transport of non-household construction, demolition and renovation waste	RON 99.83/m ³ , excluding VAT
Other material recovery outside recycling for soil, stones and C&D waste	RON 44.64/m ³ , excluding VAT
Special “Super Sack” collection and transport service	RON 246.65/bag, excluding VAT
Rental of 20 m ³ press-container	RON 1,745.52/month, excluding VAT
Rental of 12 m ³ container	RON 10.54/day, excluding VAT

FOCŞANI MUNICIPAL TARIFFS

Sa Focşani Local Council Decision No. 254 of 7 August 2024 approved revised tariffs for waste-management services.

SERVICE CATEGORY	EXCLUDING VAT	INCLUDING VAT
Construction material/rubble deposited at the operator's facility using the beneficiary's equipment	RON 93.84/m ³	RON 72.16/t or RON 111.67/m ³
Management of waste generated through internal or external dwelling renovation	RON 136.52/t	RON 162.46/t
Same renovation-waste management service	RON 211.61/m ³	RON 251.82/m ³

MEDIAŞ MUNICIPAL TARIFFS

Sa Focşani Local Council Decision No. 254 of 7 August 2024 approved revised tariffs for waste-management services.

SERVICE CATEGORY	PUBLISHED TARIFF
Concrete, asphalt, ballast residues, bricks, tiles and non-hazardous mineral mixtures	RON 343.27/t
Mixed municipal, bulky or improperly mixed waste	RON 876.95/t

RUBBLE & C&D PUBLIC MARKET MAPPING

CITY/REGION	RESPONSIBLE OPERATOR OR AUTHORITY	WASTE STREAM	PUBLISHED TARIFF
Iaşi	Salubris S.A.	Household C&D collection and transport	RON 74.69/m³ , excluding VAT
Iaşi	Salubris S.A.	Non-household C&D collection and transport	RON 99.83/m³ , excluding VAT
Iaşi	Salubris S.A.	Other recovery outside recycling	RON 44.64/m³ , excluding VAT

Focșani	Salubrizare și Servicii Publice Focșani S.A.	Rubble delivered using beneficiary's equipment	RON 60.64/t or RON 93.84/m³ , excluding VAT
Focșani	Municipal operator	Renovation-waste management	RON 136.52/t or RON 211.61/m³ , excluding VAT
Mediaș	ECO-SAL S.A.	Clean mineral C&D	RON 343.27/t
Mediaș	ECO-SAL S.A.	Improperly mixed waste	RON 876.95/t
Craiova	Municipal sanitation framework	Integrated C&D collection, transport, disposal and traceability	RON 189.79/m³ , excluding VAT

B. GLASS MARKET MAP & MUNICIPAL SEPARATION NETWORKS

The Romanian programme must distinguish packaging glass from flat and construction glass. Bottle and jar glass may enter municipal recyclable-collection systems, while windows, laminated glass, coated glass, mirrors, wire glass, frames and sealants require separate acceptance rules and confirmed specialist destinations.

CITY/REGION	OPERATOR/FACILITY	GLASS-STREAM ROLE	PUBLISHED PRICE EVIDENCE
Iași	Salubris S.A.	Non-household recyclable collection covering paper/cardboard, metal, plastic or glass	RON 32.98/m³ collection + RON 11.25/m³ sorting = RON 44.23/m³ , excluding VAT
Iași	Municipal Collection Centre	Household glass and construction-waste acceptance	Service conditions apply
Mediaș	ECO-SAL S.A.	Separate municipal collection and C&D container services	No specific public glass-sale tariff identified
Focșani	Municipal sanitation system	Separate municipal paper, metal, plastic and glass collection	Tariff embedded in the sanitation structure
Other target municipalities	Municipal companies and intercommunity systems	Packaging-glass collection	Municipality-specific

RECYCLED GRANULATE PRICE SURVEY

A comprehensive economic survey confirms that Green Mineral's output products provide a structural financial incentive for municipal procurement tenders, driving significant cost optimization:

PRODUCT TYPE / GRADE	STANDARD QUARRY MARKET PRICE	GREEN MINERAL OFFERED PRICE	NET MUNICIPAL PROCUREMENT ADVANTAGE
Sand (0-4 mm)			10-15% Below Market Rate
Gravel (4-16 mm)			10-15% Below Market Rate
Aggregate (16-32 mm)			10-15% Below Market Rate
Road Sub-base Material			10-15% Below Market Rate
Processed Glass Cullet			Aligned with National Market Rate

13. CONVENTION SCHEME & PARASOCIAL AGREEMENTS

To successfully activate the National Circular Economy program, Green Mineral proposes the execution of binding legal frameworks (parasocial agreements) with competent Romanian state and municipal authorities. These statutory commitments secure project deployment through key structural mandates:

- [REDACTED]
- [REDACTED]
- [REDACTED]
- [REDACTED]
- [REDACTED]
- [REDACTED]
- [REDACTED]
- [REDACTED]
- 8. [REDACTED]

14. INTERNATIONAL GREEN MINERAL CHAMBER OF COMMERCE

The institutional mechanism driving international deployment is overseen by the International Green Mineral Chamber of Commerce, chaired by Mr. Nando Cocco, alongside selected Romanian institutional partners.

EUROPEAN CURRICULUM VITAE: NANDO COCCO

FULL NAME

NANDO COCCO

ADDRESS

Piazza Massimo D'Azeglio, 03012 Anagni (FR), Italy

CONTACT METRICS

Tel: +39 335 5975482 | E-mail: greenmineral@pec.emx.it / cnpropromoter@virgilio.it

PERSONAL DATA

Nationality: Italian | Date of Birth: 29 July 1960

2020 - PRESENT

Director, Green Mineral SRL

Managing international Circular Economy initiatives and infrastructure projects across South Africa, Ivory Coast, Morocco, Albania, Algeria, Pakistan, Romania, and Ukraine.

2018 - 2020

Director, Green Mineral SRL - Patent Development

Supervised research, industrial design, and patent filing. Successfully secured the industrial process patent from the Italian Ministry of Economic Development (MISE) on January 8, 2020.

2005 - 2014

Director, SOC PICK SPA(Ferentino, Italy)

Managed large-scale manufacturing operations focused on technical mortar formulation and distribution.

1998 - 2008

Director, SOC NCL GROUP SRL

Led corporate strategy for aggregate production, sales, high-elevation demolition execution, and secondary inert waste recovery plant operations.

1994 - 2008

Administrator, CIF SRL(Frosinone, Italy)

Oversaw industrial production lines for tin mills and specialized commercial panel trading.

EDUCATION

Diploma di Geometra (1982)

Istituto Tecnico di Anagni (FR), Italy.

LANGUAGES

Italian (Mother Tongue); English, French, Spanish, Portuguese (Professional Written Proficiency).

TECHNICAL SKILLS

Advanced operational mastery of mineral waste processing technology, manufacturing layout design, structural building materials, and heavy recovery machinery specialized for inert recycling sectors.

CORE MISSION

Development of ambitious, ecologically sound industrial projects across four continents. Focused on the recycling of inert waste and tire derivatives with the ultimate goal to safeguard human lives and protect global ecosystems.

ROMANIAN COLLABROTOR: ENG. NAWRAJ SAPKOTA



GREEN MINERAL SRL

Nawraj Sapkota ✓

#Environment and Sustainability #Climate Resilience#Climate Finance #MultiHazard Assessment #Anticipatory Action #Circular Economy

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15. GREEN MINERAL NETWORK & GLOBAL PARTNERS

As detailed in the supporting technical dossier, Green Mineral operates within a structured international network of institutional partners, technology providers, and equity investors in the waste-to-value sector:

COUNTRY HUB	PARTNER PROFILE & OPERATIONAL NETWORK SYNERGY
Italy (HQ)	NCL Group SRL: Operational since 1998. Serves as the primary center for technical execution, granulate validation, and heavy waste plant management frameworks. http://www.nclgroup.it/en/index.html
Alseco SRL (Italy)	Core Financial & Strategic Partner: Industrial leader in rubber, polymer, and plastic waste recovery. Acts as a co-investor across international Green Mineral infrastructure deployments. https://www.alseco.it/
Morocco & Algeria	C&D Waste Recycling Programmes: Active pipeline developments focused on localized urban aggregate recovery frameworks.
Pakistan	C&D Recycling Framework: Formally entering active operational phases for localized mineral recovery.
Romania	Active Public-Private Partnership (PPP): Strategic joint negotiations underway with <i>RER Ecologic Group SRL</i> (https://rergroup.ro) to anchor European market integration.

16. INTERNATIONAL REFERENCE BASE

The global deployment matrix confirms the scalability, financial viability, and regulatory adaptability of the zero-waste processing model across diverse socioeconomic contexts:

TARGET NATION	PROJECT STATUS	STRATEGIC HIGHLIGHTS & PROGRAM PARAMETERS
Italy	Active (1998-Present)	Foundational operations via NCL Group SRL. Built in full compliance with the Ronchi Decree and EU Directive 2008/98/CE, capturing 25+ years of continuous market success.
South Africa	Negotiation Phase	Large-scale national program: 240 fixed recycling plants linked to 772 distributed collection hubs across 273 municipalities. Projected to generate 7,297 green jobs; presented to Ministry of Forests & Environment.
Romania	PPP Negotiation	Advanced joint-venture negotiations with public-utility operator RER Ecologic Group SRL to expand Eastern European operations.
Ukraine	Proposed Roll-out	National strategic reconstruction infrastructure program encompassing a targeted 20-year operational roll-out across all oblasts.
Uganda & Cape Verde	Concept Cleared	Initial technical validation completed; detailed country-specific operational dossiers are currently undergoing finalization.

Full network certification data, technical validation sheets, and updated corporate profiles are available via: <https://www.greenmineral.it/en/news/>.

17. EU LANDFILL ENFORCEMENT, ILLEGAL DUMPING AND DIGITAL TRACEABILITY

The infringement history changes the programme's risk context. The Commission issued a reasoned opinion on 7 February 2024 after 92 of 101 relevant landfills had been closed and rehabilitated but nine remained; on 29 January 2026 it referred Romania to the Court of Justice of the European Union because those nine sites were still not rehabilitated. A circular materials programme should therefore create lawful alternatives before closure, prevent recontamination and maintain auditable routing. It must not be presented as a substitute for site-specific closure, rehabilitation and aftercare.

CONTROL POINT	MINIMUM DIGITAL RECORD
Permit / project registration	Project, address, contractor, estimated waste types and quantities.
Container booking	User, container ID, category, price quote and declared prohibited items.
Collection	Date/time, vehicle, photo, GPS and estimated/actual weight.
Plant reception	Weighbridge, EWC code, contamination grade, accepted/rejected quantity.
Processing	Batch, input mass, recovered metal/glass, aggregate fractions and residue.
Product release	Test result, batch ID, intended application, buyer and quantity.
Public works use	Site, contractor, layer/application, inspection and performance record.
Landfill-closure interface	Official site identifier, legal status, responsible authority, closure milestone, rehabilitation plan reference, alternative-service start date and post-closure recontamination checks.
Landfill-derived material	Origin, competent approval, contamination sampling, excavation batch, authorised recovery/reuse/disposal route and final destination; quarantine by default until cleared.
Illegal-dumping incident	Location, estimated quantity, composition, responsible party if known and clean-up cost.
Core audit equation: incoming mass = products + separately recovered materials + process loss/moisture adjustment + residual treatment/disposal. Landfill-derived material enters this equation only after site-specific legal approval and contamination clearance.	

PUBLIC DASHBOARD

Publish monthly tonnes collected, recycling output, residual rate, public-works use, tariff revenue, product revenue, complaints, rejected loads and illegal-dumping response. Where the programme interfaces with an officially identified landfill catchment, also report alternative-service readiness, closure milestones, recontamination incidents and approved routing of landfill-derived material. Commercially sensitive details may be aggregated, but mass balance and destination evidence should remain auditable.

OFFICIAL NATIONAL COMPLIANCE RECORD

Romania's January 2026 draft National Waste Management Plan identifies the following nine non-compliant municipal landfill locations requiring completion of closure, ecological rehabilitation or formal compliance procedures.

ID	COUNTY	LANDFILL/LOCATION	DISPOSAL CEASED	OFFICIAL IMPLEMENTATION STATUS	PRINCIPAL ENVIRONMENTAL AND FISCAL CONCERN
1	Sibiu	Smârd-Mediaș	2010	Closure process delayed by administrative and property-transfer issues; estimated completion indicated for 2030	Prolonged environmental liability, site-security cost and delayed access to rehabilitation financing
2	Suceava	Hurghiș-Câmpulung Moldovenesc	2011	Feasibility study under revision; closure works planned through the Sustainable Development Programme	Continued rehabilitation and post-closure monitoring liability
3	Botoșani	Darabani	2014	Feasibility study being revised; works expected to begin under the 2021–2027 programme	Capital expenditure, procurement and implementation-delay risk
4	Botoșani	Săveni	2016	Feasibility study being revised; closure investment under preparation	Environmental protection and long-term municipal liability
5	Hunedoara	Vulcan	2016	Included among six sites prepared for closure works	Mountain-area runoff, slope-stability and rehabilitation-cost risk
6	Arad	Lipova	2017	Feasibility and closure documentation under preparation	Delayed closure, land restoration and monitoring obligations

7	Braşov	Rupea	2017	Included in the planned closure-financing package	Long-term environmental monitoring and public-budget exposure
8	Satu Mare	Satu Mare landfill	2010	Closure and ecological works reported completed in June 2025; formal reception was expected in December 2025	Final acceptance, confirmation of compliance and post-closure monitoring
9	Vâlcea	Horezu	2017	Works blocked by unresolved property and responsibility issues; estimated completion indicated for 2030	Unclear liability allocation, financing delay and continued environmental exposure

Source Note: The locations, cessation years and administrative status are taken from the revised national waste management plan.

18. ANNEXURES & DOCUMENTARY REFERENCES

The following technical, legal, and financial instruments are formally attached to this National Dossier or are available upon official request through the Executive Secretariat:

1. **Project Dossier:** Institutional framework submitted directly to the Italian Ministry of Environment (Protocol No. MCE211993, issued 18 November 2021).
2. **Alseco SRL Commitment:** Official Expression of Interest and international capital allocation pledge.
3. **EU Statutory Compliance:** Comprehensive integration text of European Waste Framework Directive 2008/98/CE.
4. **Recycled Aggregate Price Survey:** Regional price surveys and market matrices covering Kyiv, Kharkiv, Dnipro, and Lviv benchmarks.
5. **Financial Projection Model:** Summary of Revenues and multi-tier 20-year financial model encompassing Phase 1, 2, and 3 regional developments.
6. **Technical Operations PowerPoint:** Processing plant layouts, multi-stage flow charts, mass balances, and output granulate particle size specifications.
7. **Green Mineral Founder Curriculum:** Full international industrial resume of Chairman Nando Cocco.
8. **MISE Process Patent Certificate:** Italian Ministry of Economic Development industrial process patent for Zero Process Waste (Patent No. 102017000077948).
9. **NCL Group Trade Registry:** Certified Chamber of Commerce and anti-mafia governance documentation.
10. **Green Mineral Network Portfolio:** Global circular economy project blueprints across active country hubs.
11. **Convention Scheme Draft:** Parasocial agreement template configured for regional Romanian municipal authorities.
12. **Engineering Plant Layouts:** Technical schematic drawings detailing typical 3-phase plant configurations.

GREEN MINERAL ROMANIA

"Working for sustainable development and national recovery"

Green Mineral S.r.l. — Global Headquarters (Italy)

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